Doping Stability Report -- Mg in KCoO2 at 200 C

Generated: 2026-05-30 18:17   Host: KCoO2   Dopant: Mg

Executive Summary

Mg preferentially occupies the Co layer. The formation energy difference between the two layers is +2.899 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is -0.00052 A^2/ps (stable), compared to -0.00236 A^2/ps on the K site. The K-site E_f is approximate due to a charge surplus of +1 that cannot be modelled in CHGNet.

Site	E_f (eV)	Charge	Compensation	MSD pass	Coord pass	Verdict
Co	-2.553	-1	No	PASS	PASS	STABLE
K	+0.346	+1	Yes	PASS	FAIL	STRUCTURAL COLLAPSE

Mg at the Co layer -- STABLE

Charge situation

Mg replaces Co with a charge mismatch of -1. To restore charge neutrality, 1 Na atom(s) were removed from the supercell (furthest from the dopant site). The formation energy accounts for the Na-vacancy term.

Formation Energy (thermodynamic stability)

Formation energy E_f	**-2.5530 eV**	PASS	*< 1.0 eV to pass*

E_f = -2.553 eV. This value is strongly negative, indicating that Mg incorporation is thermodynamically very favorable at the Co site.

Molecular Dynamics Stability (200 C, 25 ps)

MSD slope (late-time)	**-0.00052 A^2/ps**	PASS	*< 0.005 to pass (plateau = stable)*
Final MSD	**0.045 A^2**	----	
Max displacement	**0.613 A**	----	
Coordination (min/mean/max)	**4 / 5.0 / 5**	PASS	*relaxed = 5, tolerance +/-1*
Mean bond length (MD vs relaxed)	**2.071 A (relaxed: 2.0**	----	
Lattice volume change	**+0.00 %**	PASS	*< +/-5% to pass*

Late-time MSD slope = -0.00052 A^2/ps. The negative slope is a flat plateau -- essentially no migration detected. Final MSD = 0.045 A^2; maximum displacement from starting position = 0.61 A.

Coordination fluctuated between 4 and 5 (mean 5.0), within the +/-1 tolerance of the relaxed value (5). This is consistent with normal thermal vibration. Mean nearest-neighbour distance during MD = 2.071 A.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STABLE

Mg is thermodynamically favorable and kinetically stable on the Co site at 200 C. All four stability criteria pass: formation energy is favorable, the dopant does not migrate during MD, its coordination environment is preserved, and the host lattice volume is unchanged.

? WARNING: Charge deficit (-1) cannot be compensated in CHGNet (would require adding K interstitials or oxidising Co). Running single substitution without compensation. E_f is approximate.

Mg at the K layer -- STRUCTURAL COLLAPSE

Charge situation

Mg replaces K with a charge surplus of +1. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

Formation Energy (thermodynamic stability)

Formation energy E_f	**+0.3464 eV**	**PASS**	*< 1.0 eV to pass*

E_f = +0.346 eV (approximate -- charge surplus not modelled in CHGNet). This value is small and positive, indicating that Mg incorporation is marginally feasible under equilibrium conditions at the K site.

Molecular Dynamics Stability (200 C, 25 ps)

MSD slope (late-time)	**-0.00236 A^2/ps**	**PASS**	*< 0.005 to pass (plateau = stable)*
Final MSD	**0.319 A^2**	----	
Max displacement	**0.934 A**	----	
Coordination (min/mean/max)	**3 / 4.7 / 5**	**FAIL**	*relaxed = 5, tolerance +/-1*
Mean bond length (MD vs relaxed)	**2.123 A (relaxed: 2.1**	----	
Lattice volume change	**+0.00 %**	**PASS**	*< +/-5% to pass*

Late-time MSD slope = -0.00236 A^2/ps. The negative slope is a flat plateau -- essentially no migration detected. Final MSD = 0.319 A^2; maximum displacement from starting position = 0.93 A.

Coordination fluctuated between 3 and 5 (mean 4.7), deviating by more than +/-1 from the relaxed value (5). This may indicate structural distortion, or that the bond-length cutoff is too close to the actual bond length -- check the coordination_cutoff_A setting. Mean nearest-neighbour distance during MD = 2.123 A.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STRUCTURAL COLLAPSE

The coordination criterion failed while MSD and volume both pass, suggesting a possible cutoff artifact rather than a true structural collapse. The Mg atom barely moves (MSD = 0.319 A^2) but the coordination count fluctuates because the bond length at the K site is close to the analysis cutoff. Consider increasing coordination_cutoff_A and re-running analysis.

Site Preference Comparison

The Co site is preferred by 2.899 eV over the K site. Formation energy: -2.553 eV (Co) vs +0.346 eV (K).

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Caveats and Limitations

? **CHGNet is charge-neutral.**

The potential does not model oxidation states, polaron formation, or electron/hole localisation. Formation energies for aliovalent dopants (charge mismatch != 0) are approximate even when structural compensation defects are included.

? **No Freysoldt/Kumagai correction.**

Electrostatic finite-size corrections for charged defects require the host dielectric tensor and the DFT electrostatic potential, neither of which is available from a neural-network potential. For publication-quality E_f of charged defects, follow up with DFT using doped or pymatgen-analysis-defects.

? **Metal-rich reference state.**

Chemical potentials are computed from elemental ground-state structures. Synthesis under oxide-rich or other conditions shifts all E_f values by additive constants; the relative ordering of sites is unaffected.

? **MD timescales are short (25 ps default).**

This is sufficient to assess local site stability and rapid migration, but cannot resolve slow diffusion mechanisms. A non-plateauing MSD confirms instability; a plateau does not guarantee stability on longer timescales.

? **Energy rankings are more trustworthy than absolute values.**

CHGNet is a universal MLIP trained on r2SCAN DFT. Formation-energy errors for transition-metal oxides are typically 50-150 meV/atom. Use relative orderings (which site is preferred?) as the primary result.